Technology, society, environment & safety

GRAPHICS WITH MR KELLY

National 5 · CAD & technology · Worksheet

Name _____ Class _____ Date _____

By the end of this worksheet I can…

- ☐ explain the paperless office, remote working and communication across international boundaries
- ☐ describe ways to reduce the environmental impact of printing and packaging, including soy and wax inks
- ☐ describe safe working practices

I DO — Worked example

Read how a full-mark answer is built

A design company lets staff work from home and meet by video call.

1 Describe two benefits of remote working. (2 marks)

Points that earn marks from the marking instructions: one ✓ per point

- ✓ Less travel time and cost
- ✓ Less travel means less pollution from vehicles

Why marks are not awarded

- ✗ "Reduces pollution" without saying how

WE DO — Guided practice

Use the starters and word bank to help you

2 Explain why soy-based ink is better for the environment than petroleum-based ink. (2 marks)

Sentence starters

One reason is … because …

Another reason is … so …

Word bank

recycled · emissions · harmful · source · easier · renewable · remove · chemicals

Tip: 2 marks = 2 separate, developed points.

YOU DO **On your own** No notes. Check your answers afterwards.

3 Quick check: tick one box for each.

a) Which answer scores for an environmental benefit of remote working?
☐ It reduces pollution and is better for everyone ☐ Less travel to work means less pollution from vehicles ☐ It is better for the environment overall ☐ People are happier and more productive at home

b) Soy ink is more environmentally friendly because…
☐ It comes from a renewable source and is easier to remove when recycling ☐ It is cheaper than any other type of printing ink ☐ It dries instantly, so the presses use less energy ☐ It gives much brighter colours than normal inks

c) Fitting nets closely together on a sheet…
☐ Wastes card ☐ Makes printing slower ☐ Reduces waste ☐ Is unsafe

d) Which reduces eye strain?
☐ Working without breaks to finish sooner ☐ Turning the screen brightness right up ☐ Using smaller fonts on the screen ☐ Regular breaks from the screen

4 Key terms: write the letter of the matching description next to each term.

1. **Soy ink**	____	**A** Vegetable-based, renewable ink; easier to remove when recycling paper.
2. **Wax ink**	____	**B** Saves paper, ink and storage; easier sharing and searching.
3. **Paperless office benefit**	____	**C** Solid ink blocks: no cartridges, less packaging waste.

5 Describe one benefit of remote working. (2 marks)

6 Explain what is meant by ‘Safe screen work’. (2 marks)

Check and reflect
Mark your answers with the answer sheet. Go back to the checklist on the site and re-rate each “I can” statement above. Which one still needs work?